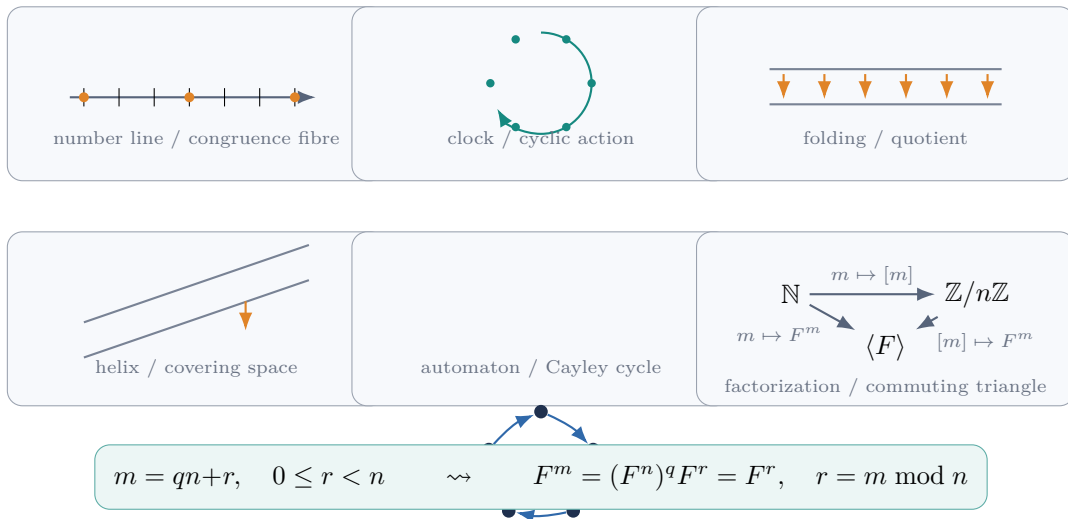


Modulo Normalization: a Visual Atlas

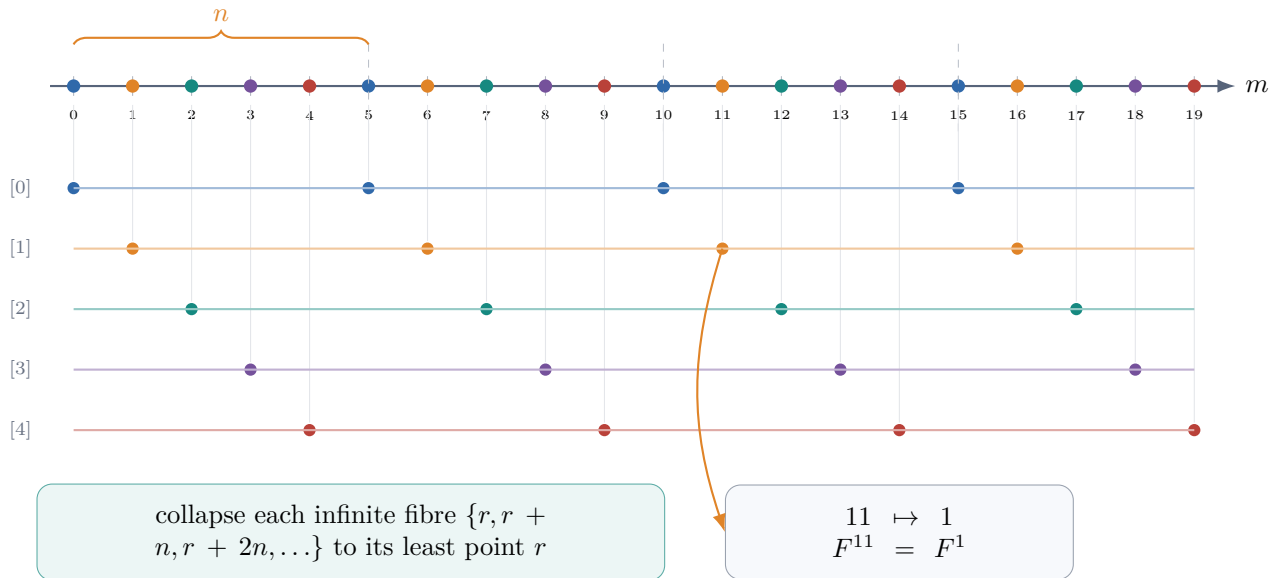
One identity, many pictures: $F^n = 1 \implies F^m = F^{m \bmod n}$



Every plate below redraws this same normalization pattern.

1. Number Lines as Congruence Fibres

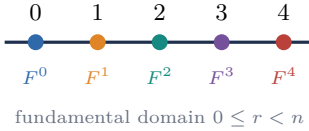
$n = 5$: vertical colour means equal Frobenius power



$m \bmod n$ is a coordinate on the quotient, not merely a shorter exponent.

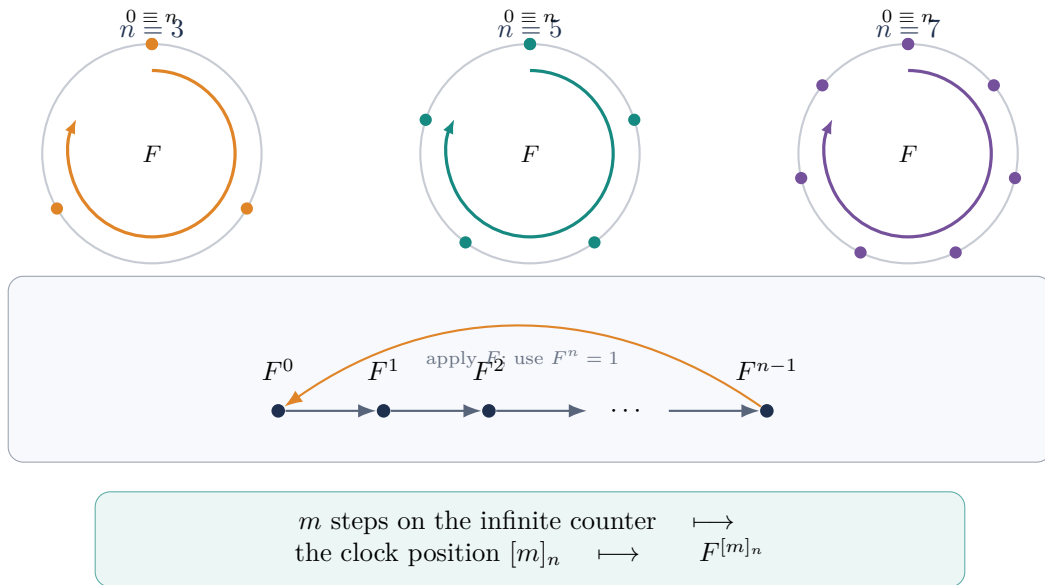
Cut into periods; stack; project to the fundamental domain

d quotient q



3. Frobenius Clocks

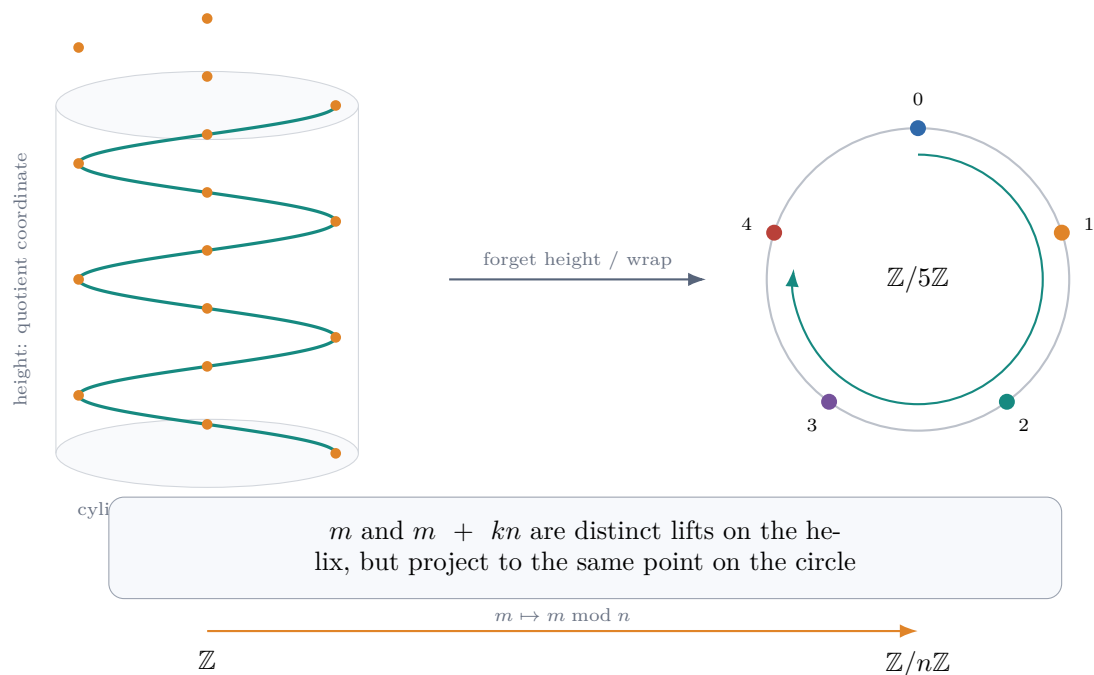
Iteration moves one tick; the n th tick returns to the identity



The clock is the Cayley graph of the cyclic action generated by Frobenius.

4. Helix, Cylinder, and Covering Map

$\mathbb{Z}$ unwinds the cycle; modulo n wraps it back



Normalization chooses the distinguished lift $0, 1, \dots, n - 1$.

5. Combinatorial Faces of the Same Quotient

Beads, words, states, and orbits

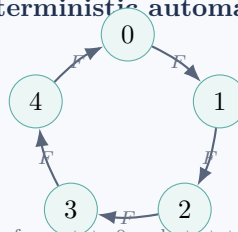
periodic word

primitive block



position m reads symbol $m \bmod 5$

deterministic automaton



m inputs from state 0 end at state $m \bmod 5$

orbit partition

0, $0 + 5$, $0 + 10$, $0 + 15, \dots$

1, $1 + 5$, $1 + 10$, $1 + 15, \dots$

2, $2 + 5$, $2 + 10$, $2 + 15, \dots$

3, $3 + 5$, $3 + 10$, $3 + 15, \dots$

4, $4 + 5$, $4 + 10$, $4 + 15, \dots$

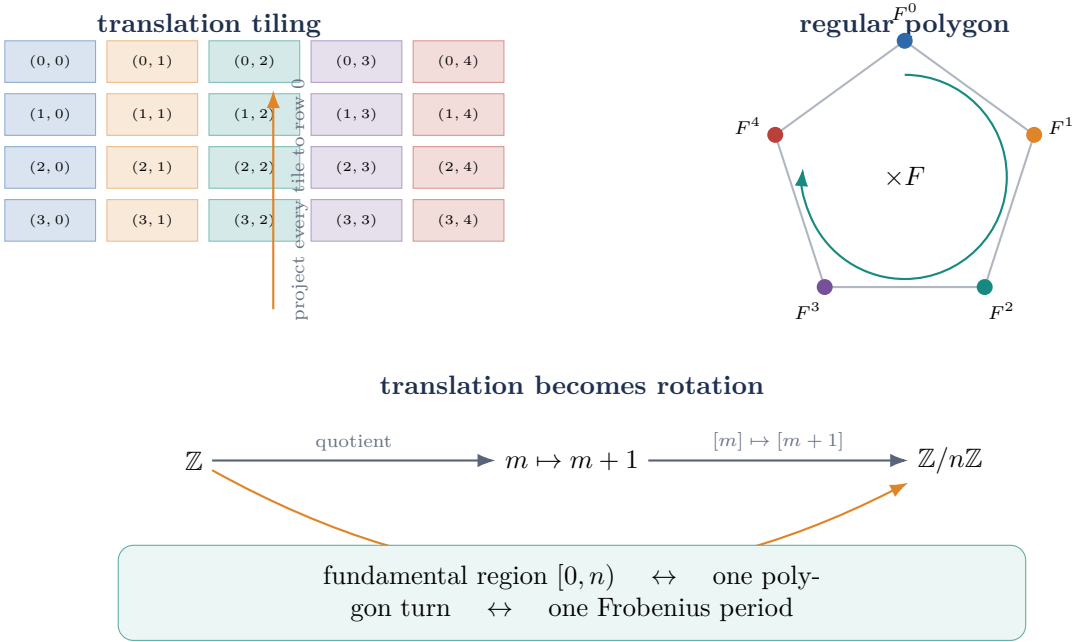
token cancellation



$$m = qn + r$$

6. Geometric Normal Forms

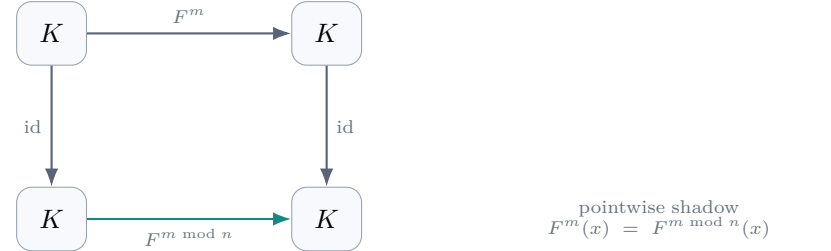
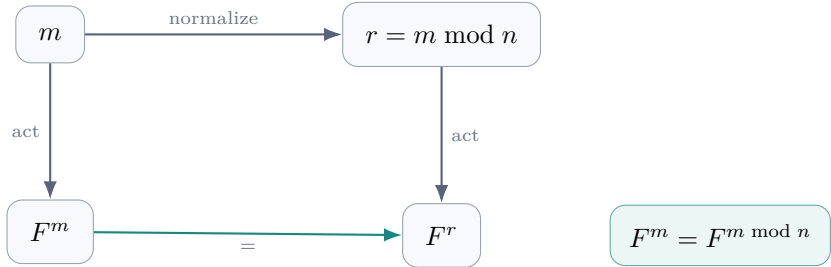
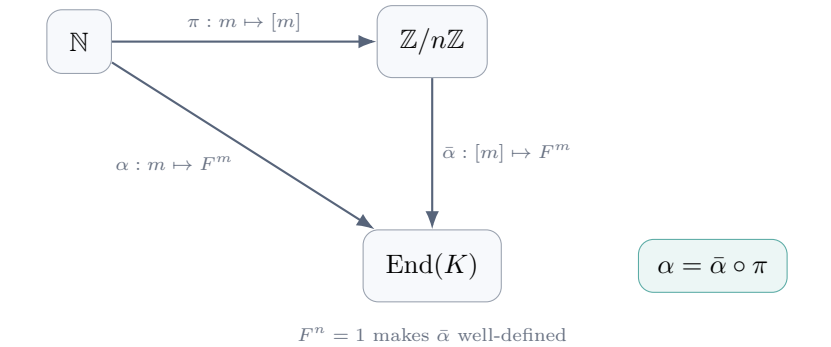
Translations, tessellations, and rotations



The modulo representative is the point inside a chosen fundamental region.

7. Commutative Diagrams

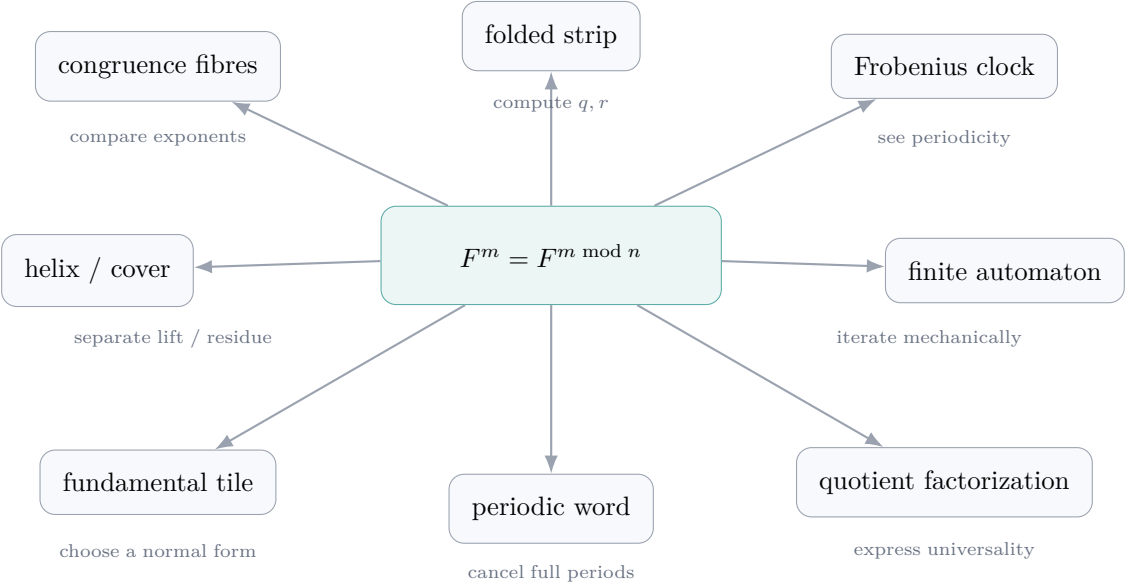
The action factors uniquely through the quotient



Categorically: a periodic monoid action descends along a quotient.

8. One Pattern, Nine Reading Directions

Choose the picture that exposes the next proof step



$$m = qn + r \quad F^n = 1 \quad F^m = (F^n)^q F^r = F^r$$

$r = m \bmod n$ is the canonical residue, state, tick, tile, lift, symbol, and quotient coordinate.